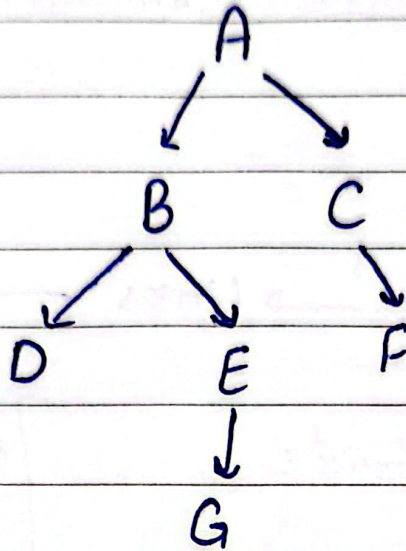


AI Lab:

Graph:



BFS Trace:

Queue	Path
[A]	A
[B, C]	A → B
[C, D, E]	A → C
[D, E, F]	A → B → D
[E, F]	A → B → E
[F, G]	A → C → F
[G]	A → B → E → G

Processed node:

A, B, C, D, E, F, G

Final Path:

$A \rightarrow B \rightarrow E \rightarrow G$

DFS Trace:

Stack	Processed node	Path
[A]	A	A
[C, B]	B	$A \rightarrow B$
[C, E, D]	D	$A \rightarrow B \rightarrow D$
[C, E]	E	$A \rightarrow B \rightarrow E$
[C, G]	G	$A \rightarrow B \rightarrow E \rightarrow G$

DFS visit order:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow G$

Final Path:

$A \rightarrow B \rightarrow E \rightarrow G$

Verification:

The BFS and DFS Traces were verified against the given code and both algorithm produce the final path $A \rightarrow B \rightarrow E \rightarrow G$.

2-DLS Result:

$A \rightarrow B \rightarrow E \rightarrow G$

depth counting:

$A = 0$

$B = 1$

$E = 2$

$G = 3$

DLS
Limit

Status

Path

1

Cut off

—

2

cut off

—

3

found

$A \rightarrow B \rightarrow E \rightarrow G$

4

found

$A \rightarrow B \rightarrow E \rightarrow G$

3-IDDFS Results:

Maximum
depth

Status

Path

Limits

1

Failure

—

0, 1

2

Failure

—

0, 1, 2

3

Found

$A \rightarrow B \rightarrow E \rightarrow G$

0, 1, 2, 3

4

found

$A \rightarrow B \rightarrow E \rightarrow G$

0, 1, 2, 3

Important:

In IDDEs, once the goal is found at depth 3, it stops. Therefore, when maximum depth is 4, limit 4 is not actually searched.

4-Why limit below 3 fails:

The goal G is at depth 3.

Limit 1: Search can only reach $A \rightarrow B$ or $A \rightarrow C$.

Limit 2: Search can reach E , but not its child G .

Limit 3: Search can reach G , so the goal is found.

Therefore,

Limit 1 & 2 cannot succeed because they do not allow the search to reach the goal at depth 3.

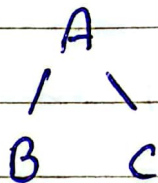
5-Repeated work in IDDFs:

IDDFs repeatedly searches the upper part of the tree.

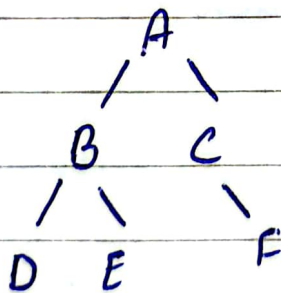
Limit 0:

A

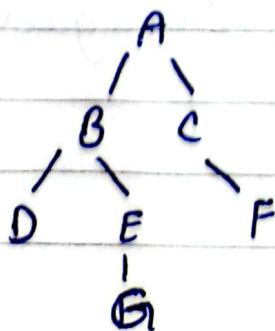
Limit 1:



Limit 2:



Limit 3:



So, nodes A, B, C, D and E maybe examine again at successive depth limit. The repeated work is the main disadvantage of IDDFs. However, it allows IDDFs to combine the low memory usage of DFs with the shallowest. Solution property of BFs when the step cost are equal.